DESIGN OPTIMIZATION AND OPTIMAL CONTROL STRATEGY OF THREE PHASE GRID TIED PHOTOVOLTAIC WITH DIESEL GENERATOR BACKUP SYSTEM

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Declaration

I declare that this is my original work and has not been presented for a degree in any other university.

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Contents

PV Photovoltaic

GHG Greenhouse Gas

AC Alternative Current

DC Direct Current

DG Distributed Generators

DDG Diesel Driven Generator

SG Synchronous Generator

PI Proportional Integral

MIMO Multi Input Multi Output

MPC Model Predictive Control

OMPC Optimal Model Predictive Control

AVG Average voltage Control

FIT Feed In Tariff

BESS Battery Energy Storage System

MPPT Maximum Power Point Tracking

PLL Phasor Locked Loop

GA Genetic Algorithm

UPS Uninterrupted power supply

TOU Time of use

PCC Point of Common Coupling

Chapter 1

INTRODUCTION

1.1 Background

The demand of electrical energy is increasing day by day due to population growth, global economic growth, and rapid industrialization . As a result, all over the world, power system networks are being operated in stressed conditions to meet this energy demand [1, 2]. Kenya's electricity sector currently experiences many challenges such as : periodic and frequent outages, low access rate, and inability to meet the growing energy demand. Kenya's vision 2030 has therefore identified energy as a key foundation upon which the economic, social, and political pillars of the long-term development strategy will be built [3, 4]. This high rate of energy consumption necessites a great investment in generation, reliable, and efficient power supply to avoid energy crisis and the blackout [5]. Therefore, Kenya has liberalized energy sector in order to exploit other sources of energy generation to reduce the stress on the main grid [3, 4].

In fact, fossil fuels represent the major energy resources in most countries, but these resources emit greenhouse gas (GHG) that highly affect the environment and human health [10]. Besides, the installation of diesel driven generator (DDG) is the most substitution for the primary source of energy during backout period. However, this solution causes massive air pollution, requires regular maintenance and is highly influenced by fuel consumption. Despite their various disadvantages, they are still being integrated in distribution systems. Renewable energy is then adopted in Kenya in order to reduce the dependency of fossil-fuel. The ability of exploiting renewable energy resources provides a promising future for power generation to resolve future energy requirement and lower the greenhouse gas emissions (GHG) in Kenya. Furthermore, Kenya lies along Equator with abundant sunshine all over the year, it is endowed with very abundant solar energy that remained unexplored and unexploited for altenative energy solutions [7]. Photovoltaic energy is a promising energy sources because of its advantages such as low cost, clean, available, maintenance-free and so on. The integration of solar energy in the distribution power system can therefore change the single power into bidirectional power flow [4, 5, 6, 10].

A typical microgrid (MG) contains different kinds of distributed generators (DGs). Accordingly, microgrids can be self-sustainable source of energy, hence is the best option for protecting environment as well as solution towards the limited availability of fossil fuel [5, 6, 10]. However, the supply of photovoltaic (PV) energy can be extremely intermittent and unreliable, hence, they require effective use of some interfaces for high performance and reliability to mitigate these uncertainties and disturbances [5, 10, 11]. The intermittent effects of solar energy source can cause large fluctuation of power quality and energy flow. Since microgrids contain single and three-phase types of loads, using three phase is a standard issues nowadays due to numerous reasons such as zero neutral currents and power constancy. The overall system has the tendency to increase in load unbalanced due to loads on single lines increase, lack of planning and unbalanced three phase loads. Moreover, non-linear and unbalanced loads could further affect the reactive power sharing whilst regulating the active power, and it is difficult to share the reactive power accurately and this can lead to unstable microgrid.

The application of hybrid photovoltaic (PV)- diesel driven generator (DDG) systems have been widely recognized as a reliable, feasible, and environment friendly solution to supply power to remote locations when it is well designed and managed during their operations. The coordinated operation of the Photovoltaic- diesel driven generator (DDG) can offer a chance to eliminate the need for energy storage device, dumping load for improvement of the system economics. Besides, the waste battery energy storage sytems (BESS) may cause energy losses and environnement pollution in that they are realesed in nature [14, 16].

In fact, Current technologies of three-phase grid connected photovoltaic-diesel generator without baterry bank are not completely performant, reliable and efficient. This system in medium or large scale and connecting it into a utility grid always introduce technological challenges related to optimal control operation, system design and system sizing [2, 16]. To improve the reliability, performance and energy power supply of overall system, efficient technologies are thus required. Many researches are required for three phase grid connected photovoltaic (PV)-diesel driven generator hybrid system such as Maximum Power Point Tracking (MPPT), improvement of the synchronization of photovoltaic to the utility grid and control of voltage source inverter. Moreover, there is necessity to develop the optimal design in order to obtain the minimum life cycle cost, reduce greenhouse gas emissions, enhance reliability and optimal control strategy to maintain the energy balance, cost-savings and power reliability. Furthermore, this system can also allow the end-user to feed its own load utilizing the available solar energy, and the surplus energy can be fed into the grid through a suitable feed-in-tariff [2, 12].

Besides, the design optimization is carried out by genetic algorithm and the overall system has to operate in its optimal conditions to guarantee optimal operation by model predictive control. Model predictive control (MPC) is chosen due to the several benefits that it can provide such as fast tracking response and simple inclusion of system non-linearities and constraints in the controller. Model predictive control (MPC) considers the system model for predicting its future behaviour and determining the best control action on the basis of a cost function minimization procedure [27].

1.2 Problem Statement

Since there is increase demand of energy nowadays, many developing contries have been struggling with energy crisis. Therefore, power systems are being operated in stressed conditions for insufficient energy supply, this lead to system operating to its limit of stabiliy and periodic outage. Unreliable supplies of electricity by main grid utilities have forced many users, from households to large enterprises, to rely on diesel driven generators as backup. Besides, diesel generators emit greenhouse gas emissions and are not cost-efficive due to fuel and maintenance cost. To overcome this issue of energy crisis, greenhouse gas emissions and stressed power system, distributed generated (DGs) sources for penetration of renewable energy is thus adopted. Ditributed generation has many advantages such as, closeness to the customers, increased reliability and reduced transmission loss [16]. End-users invest in distribution generation in order to generate their own electricity [20]. However, the penetration of renewable energy produce serious problems in terms of dynamic power fluactuation that can lead to unstable microgrid due to sudden load changes and intermittent nature of renewable energy. It is well known that the power supply is unreliable in kenya, thus inadequate power supplies impose heavy losses in economy on the private sector. Frequent power outages mean serious losses in forgone economy and damaged equipment. Thus, kenya vision 2030 identifies energy as a main foundation upon which the development strategy will be built [7]. Electricity access in Kenya is low despite the governments ambitious target to increase electricity connectivity from the current 15% to at least 65% by the year 2030. Therefore, to meet energy future demand, renewable energy need to be increased in the coming years [5]. However, power generation in Kenya is mainly from hydro, geothermal, diesel, gas and wind. Kenya should also harness photovoltaic for its great availability throughout all year [7].

Photovoltaic is one of rapidly growing energy due to its low cost, greatest availability, maintenance-free and so on. In contrast, one of the main disadvantages with photovoltaic energy is not reliable in terms of sustainability and power quality due to their intermittent nature and discontinuity in delivery power. Due to unreliable power supplier, three-phase grid-tied photovoltaic (PV) with backup diesel driven generator systems can ensure uninterrupted power supply (UPC) at reduced cost without violating environment constraints. As matter of fact, they are becoming popular to supply continuously energy to the loads because the application of hybrid photovoltaic-diesel driven generator has been widely recognized feasible, cost-effective and environment friendly in remote areas [17]. Furthermore, battery energy storage system is not either environmental nor economical friendly solution and its lifetime is very limited only 5 years.

Grid connected photovoltaic (PV) with diesel driven generator (DDG) backup systems without battery bank play an important role in the production of distributed energy. This system still has problems of design optimization and operational efficiency. To tackle these problems, the system requires efficient technologies. It is very important to establish an optimal design and control strategy which enable to solve the dispatching problem of the entire hybrid system in order to guarantee reliability, superior performances and reduce operation cost [1, 9, 14].

In other words, the integration of photovoltaic into the main grid with diesel driven generator backup without storage system rises complex challenges such as optimal design and operational efficiency. Thus, There is need to develop design optimization and optimal control strategy to improve the reliability and efficiency by minimizing overall cost and greenhouse emissions.

1.3 Justification of Study

In order to meet future energy demand, reduce the pressure on the main grid and improve the reliability of system, distributed generations (GDs) sources of energy are needed [5, 6]. Kenya government intends to increase electricity access rate to 65 % by 2030 to meet future energy requirements [3, 4]. However, the most significant difficulties in generating electrical energy using fossil fuel resources are their high cost, shortage and negative environmental impacts. Thus, renewable energy sources such as solar, hydro, wind, among others, are the most promising in resolving the future energy crisis, reduce stresses on the main grid and greenhouse emissions. Among all these renewable energy, photovoltaic energy is one of the most developed renewable technologies nowadays [5, 6]. Kenya has the potential of generating electricity from solar energy than it is consumed each year from its national grid

[3, 4, 7].

Grid connected photovoltaic system operate in parallel with main grid and as result do not require batteries storage system. To ensure continuity of power supply with cost-effective, the system require the diesel generator in case of outage of the main grid [17]. Three-phase grid connected Photovoltaic -diesel hybrid systems are increasing in several developing countries for its cost-effective. They have the following benefits:

- Eliminate the need for energy storage as well as investment, replacement costs and the energy losses associated to battery energy storage system (BESS);
- Feed surplus of energy into the utility grid, thus economic profit can be achieved;
- Reduce power drawn from the utility grid power system;
- Improve the voltage profile in the distribution lines, thereby reducing transmission and distribution losses;
- Reduce the dependency of fossil fuel generator;
- Assure permanent supply of energy to the loads;
- Reduce total production costs;
- Lower greenhouse gas emissions (GHG), etc

1.4 Objective

The main objective is to develop an optimal design and control strategy of three phase grid-tied photovoltaic-diesel generator hybrid in order to minimize the investment,operating cost, greenhouse emissions, and improve reliability while satisfying the operational constraints, unbalanced loads. To achieve this objective, the following specific objectives must be accomplished:

- To optimally designed three-phase grid connected photovoltaic-diesel hybrid system without storage battery bank based on multi-objective such that reducing life cycle cost, maximizing reliability and lowering greenhouse gas emissions by using Genetic Algorithms.
- To develop a novel optimal control algorithm using model predictive control to manage and optimize the energy in the entire system by considering firstly maximum power point tracking (MPPT) and secondly not consider the maximum power point tracking.
- To evaluate the payback period of the entire system owing to the appropriated feed-in tariff in Kenya leading to the huge revenue from photovoltaic energy sales.

1.5 Scope

An important connection of photovoltaic into the main grid with diesel driven generator as backup is growing rapidly in order to exploit solar source of energy to mitigate future energy demand, and to ensure continuous power supply when the utility grid cuts off. More specifically, the scope of this work is focused on design optimization and optimal control strategy of three-phase grid connected photovoltaic with diesel driven generator backup capable of compensating unbalanced loads and considering relevant constraints. Thus, this is done by decoupling the active and reactive power within the entire system. Moreover, the simulation is carried out firstly not considering maximum power point tracking (MPPT) and secondly by considering the most commonly used maximum power point tracking (MPPT) based on Perturb-Observe. In order to reach the objectives, the load profile of Jomo Kenyatta University of Technology (JKUAT) is considered in this work. Additionally, an economic analyse is carried out to determine the most suitable solution that can be implemented at Jomo Kenyatta University of Technology (JKUAT).

Chapter 2

Literature review

Before developing the optimal design and control strategy applied in this research work, it is vital to examine various studies that have been proposed in literature. The aim goal of this chapter is to identify areas that are over-studied, as well as gaps in order to avoid duplication.

2.1 Introduction

Electrical energy plays an important role in our daily life activities due to social, economic and industrial growth. To meet future energy requirements, alternative energy such as renewable energy which is friendly to the environment is needed for reliable power system. Therefore, three phase grid tied-photovoltaic with diesel back up energy system is taking the biggest share to enhance energy reliability and reduce greenhouse gas emission [1, 2]. This chapter is devoted to a review of relevant researches done by other scholars in other part of the world related to three-phase grid connected photovoltaic with diesel driven generator backup.

2.2 Grid connected photovoltaic-Diesel hybrid system

Most of problems tie on optimal design, sizing and technico-economic analys have employed software such as hybrid optimization model electric renewable (HOMER) to simulate hybrid systems. However, there is few research in the design of optimal control strategy for higher control, supervisory level and real-time control techniques in dynamic scheduling. Many studies presented the operation and control of hybrid systems under normal operating conditions [15, 16, 17].

In [15], the authors deal with the cost optimization of electricity generation from a grid connected hybrid solar and diesel generator using software packages Hybrid Optimization Model Electric Renewable (HOMER). The proposed hybrid system is made of the grid connected photovoltaic (PV) that supplies the full load with the exception of the air-condition systems while the national grid is used to cover the rest. Hybrid optimization model electric renewable (HOMER) software is not appropriated when it comes to consider the dynamic scheduling of the system. Besides, Hybrid optimization model electric renewable (HOMER) can only address single objective function for minimizing the net present cost while the multi-objectives problem cannot be addressed. Dealing with multi-objective cost optimization Generic Algorithm (GA) is provided best result. Genetic Algorithm (GA) can be the most favoured method for single objective and multi-objective optimizations of hybrid energy systems such as minimizing life cycle cost and greenhouse emissions and enhance reliability.

Several researches have shown that large range of dynamic modes can be observed in microgrid, thus dynamic scheduling and supervision are necessary. Using Hybrid optimization model electric renewable, the authors do not consider these dynamic behaviours of the system that depend on disturbances and on changes in the operating point. The dynamic behaviours, constraints and the control strategy for enhancing performances optimization and multi-objective function cannot be tackled by hybrid optimization model electric renewable (HOMER). Besides, the quality of generated electricity is dependent of the outputs. Hybrid optimization model electric renewable (HOMER) is very limited, it cannot be used to optimize the design of renewable energy systems in microgrid where the design of some renewable energy system affects electric consumption of other energy systems.

In [16], the authors proposed a seamless control methodology photovoltaic diesel to operate both in grid connected and isolated mode. The proposed control scheme does not require any islanding detection mechanism to alter the control philosophy from grid connected to autonomous mode. This seamless control structure has been able to tackle some of limitations of existing schemes in literature.

However, this control methodology has limitations that have not been addressed. As the power management is photovoltaic first, the maximum power point tracking has to be extracted whenever possible. There is no battery storage system or damping loads, when the micro grid gets isolated from the main grid, excess energy fed to grid can cause fluactation of active and reactive power flow that can lead to instabilty. In grid connected mode the conventional scheme strategies may provide good performance. However, in [17], authors demonstrate that by using conventional control strategies, the entire system is not flexible for soft transition between grid connected and off grid.

Besides, the control scheme has not been tested for scalability of entire hybrid power system and variability of both the solar irradiation and electrical loads. Under these conditions, the control scheme has to be modified in order to consider the changes in the system. In order to minimize the cost of energy, the optimal sharing algorithm have to be developed.

Additionally, a second order sliding mode control is used to control the voltage source inverter (VSC). Sliding mode control is a well known robust control technique for non-linear systems which guarantee the elimination of disturbances. However with this control scheme is not possible to deal with constraints and forecast possible future behaviour of controlled variable.

To track maximum power for photovoltaic side, perturb and observe algorithm has been used. However, the control scheme is capable of operating the photovoltaic (PV) array at its maximum power point tracking but the technique has its poor dynamic response and swinging around the maximum power point during steady state operation. During fast changes in atmospheric conditions, the perturb and observe technique may erroneously move in the wrong direction on the photovoltaic power curve if each perturbation caused power to increase.

Voltage source converter is the one that takes care of reactive power control resulting in an increase in the converter rating. Therefore,for good performance each source has to take care of reactive power control as well. The proposed control structure deal with single machine, therefore an extension of the control scheme to multiple machine scenario must therefore be investigated. In particular, this control strategy scheme does not consider the load unbalance, non-linear and controllable loads in order to ensure security of the system. In addition, the system has three wires therefore, the authors do not consider the case of unsymmetrical voltage which may occur due to unequal distribution of single phase loads on three-phase lines, asymmetrical transmission impedance etc. The voltage phase shifts can deviate from 120°, over voltage and under voltage can occur at different nodes of micro grid, etc.

In [17], the authors investigate on a novel neuro-fuzzy controller based on data analytics to ensure frequency stability for diesel-photovoltaic hybrid AC microgrid. The system uses LC filter to mitigate harmonic. The diesel driven generator (DDG) is controlled by a conventional control method PI controller and the voltage source converter (VSC) by sliding mode controller. The control strategy based on neuro-fuzzy can ensure smooth transition between grid connected and isolated mode. However, it has been shown that artificial intelligence control technique for photovoltaic needs much time to collect data, recalculate parameters of controllers and requires a large memory to save data acquisition [19]. In addition, for excessive generation power due to maximum power point tracking strategy, power is supplied to the diesel driven generator and it can operate as a motor. The conventional speed control scheme of the diesel driven may fail to keep the speed with accepted limits. Therefore, this will cause the instability of voltage and frequency. On the other hand, it may damage the load equipment and/or the interface inverter resulting in unwanted financial loss. This will greatly increase the tear and wear in the diesel engine that may initiate mechanical deterioration to the diesel engine. The control system must be flexible in both grid connected and standalone modes. This study reported to the control of grid connected photovoltaic under normal operating condition. The conventional three leg voltage source converter (VSC) is proved a best choice to supply three-phase balanced loads. However, the entire system does not have the ability to supply the unbalanced load. The control scheme should be able to compensate the load unbalance by the photovoltaic and diesel driven generator. Therefore,this proposed control structure can be expanded in the case of harmonics reduction and compensation unbalanced and non-linear loads using an active control of the forth wire(neural) of the inverter. In addition, the performances optimization are not invistigated in this work [17].

2.2.1 Summary

Several researches have been carried out for grid connected photovoltaic with diesel generator backup. However, most of them investigate in techno-economic analysis using hybrid optimization model electric renewable (HOMER). Only two of them [16] and [17] have considered the dynamic behaviour of the entire system. All of them use slide mode control to control the voltage source inverter and consider balanced and linear loads. As the system does not require battery bank, the maximum power point tracking designed to enhance photovoltaic array can make the system unstable when it comes to transition from grid connected to islanded mode. The design and control of entire system need further investigation in terms of performance optimization and operational efficiency in grid connected and autonomous modes. Most importantly, the design optimization and control strategy can enhance the reliability, improve energy-efficiency, lower the cost and greenhouse gas emissions. The authors should also consider to design the entire system based on compensate the unbalanced loads and system's constraints.

Chapter 3

Methodology

This research focuses on developing a novel optimal design and control strategy of three-phase grid connected photovoltaic - diesel generator system without neither battery energy storage system (BESS) nor dump loads in order to reduce the life cycle cost and green house gas and minimize the operation cost of energy flow. The structure of whole system is depicted in Figure 3.1. Genetic algorithm is used for optimally designed the system and model predictive controller (MPC) is designed to control the voltage control source (VSC).

3.0.1 Optimal design

As it has been stated above, hybrid optimization model electric renewable (HOMER) can only address single objective function for minimizing the net present cost while the multi-objectives problem cannot be addressed. Dealing with multi-objective cost optimization genetic algorithm is provided best result.

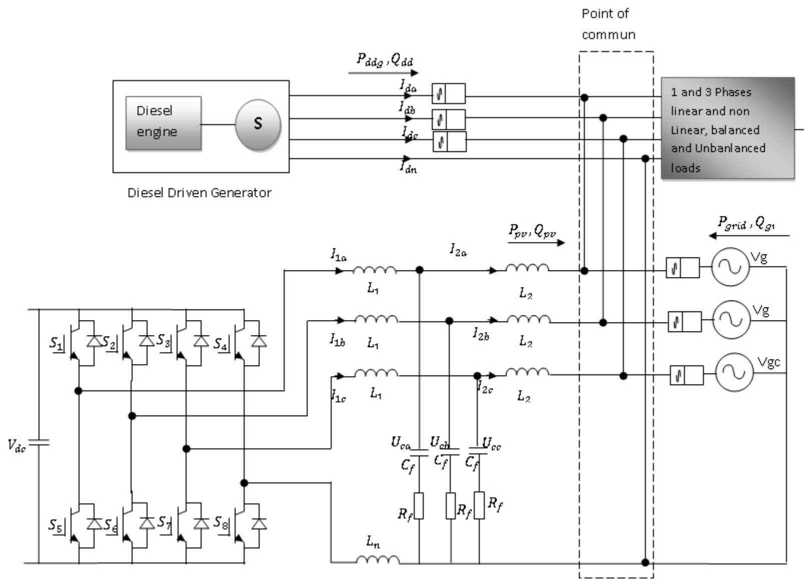

Figure 3.1: Three phase grid tied PV-DG structure

Economics and reliability

Both reliability and economic issues play important roles in the design process of microgrid. Hybrid renewable energy system is generally entail high capital costs, even though they have low operation and maintenance (O-M), thus an economic analysis is required to determine the optimum design for cost and benet ratio, and achieve the least possible unit price of the system. Therefore, an optimization technique is essential to minimize the cost while maintaining reliability. In order to optimize the parameters of the entire system a measurement to compare different configurations needs to be defined. An appropriate measure is the life cycle cost (LCC).

$$LCC = C_c + C_o + C_s \tag{3.1}$$

The reliability is also a great measure of system performance as they eliminate to a great extent, the uncertainty that comes with solar energy resources, grid and load disturbances. One parameter that can help to evaluate the reliability of the system is the loss of power supply probability (LPSP) [25]. Genetic Algorithms can be used to perform the optimal design with minimum Life cycle cost, greenhouse gas emissions and maximum reliability.

$$LPSP = \frac{\sum_k LPS_k}{\sum_k E_{Lk}} \tag{3.2}$$

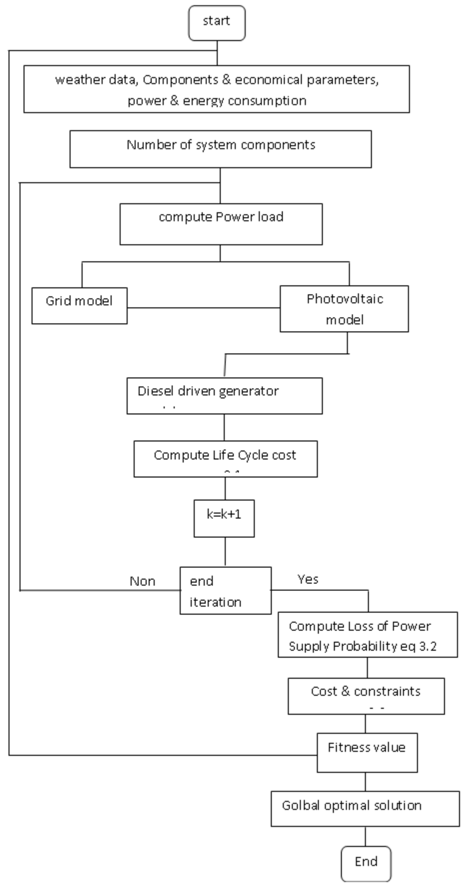

Figure 3.2: Design optimization flow cart

Where LPS_k is the deficit of power at the sampling time k. it is termed Loss of power supply. The photovoltaic-diesel generator backup design optimization is then formulated as follows

$$\begin{cases} minimize J_1 = capital + Maintenance + Replacement \\ minimize J_2 = \sum F_{DG}^{ele} \times GHG_{C0_2}^{ele} \\ subjectto LPSP = 0 \\ P_{load,k} \leq P_{max}(k) \end{cases} \tag{3.3}$$

With $P_{load,k}$ the amount of power consumed by the load at step period k, F_{DG}^{ele} is the energy consumption, and $P_{max}(k)$ the contracted power limit from the grid. The flow chart of design optimization is illustrated in figure 3.2. The Genetic Algorithm (GA) will be applied using the optimization toolbox of MATLAB.

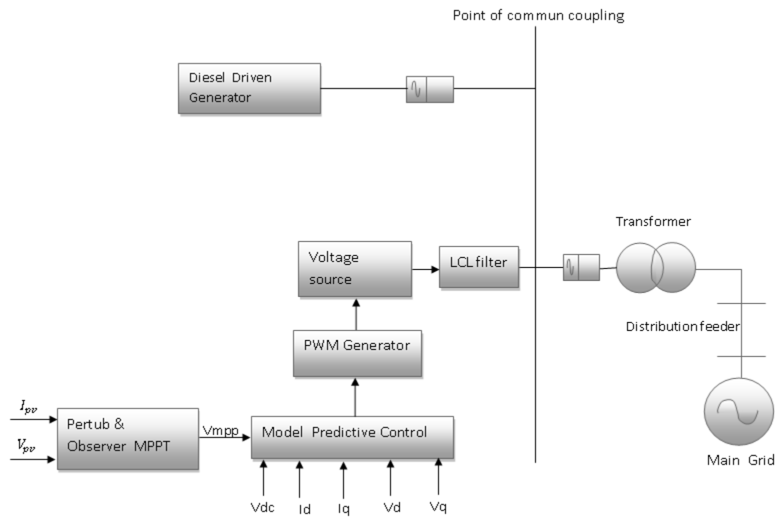

Figure 3.3: structure of the control system

3.0.2 Optimal Control

Optimal control strategy is constructed in order to manage and optimize the energy of the hybrid system. On one hand, the maximum power point tracking is considered and one the other, It is not. Figure 3.3 shows the one with maximum power point tracking (MPPT). The three-phase output voltage in the abc frame can be represented as

$$\begin{bmatrix} V_a \\ V_b \\ V_c \end{bmatrix} = \begin{bmatrix} V_{max}cos(wt) \\ V_{max}cos(wt-120) \\ V_{max}cos(wt+120) \end{bmatrix} \tag{3.4}$$

In this study, to handle three-phase grid connected photovoltaic with diesel driven generator park transformation is chosen. It uses the reference frame transformation abc to d-q to transform the grid voltage and current waveforms into a reference frame which rotates synchronously with the grid voltage. This makes control variables as direct current (DC) quantities which are easier to handle in closed loop. Every deviation in the grid voltage and current will be shown to the corresponding d-axis and q-axis components. The conversion park transformation matrix is given by:

$$T^{-1} = \frac{2}{3} \begin{bmatrix} cos(wt) & cos(wt-120) & cos(wt+120) \\ sin(wt) & sin(wt-120) & sin(wt+120) \\ \frac{1}{2} & \frac{1}{2} & \frac{1}{2} \end{bmatrix} \tag{3.5}$$

where the transformation in terms of voltage is given by

$$\begin{bmatrix} V_a & V_b & V_c \end{bmatrix}^T = T \begin{bmatrix} Vd & Vq & 0 \end{bmatrix}^T \tag{3.6}$$

The voltages and currents are defined as follows

$$V_{cap} = \begin{bmatrix} U_{ca} & Ucb & Ucc \end{bmatrix}^T \tag{3.7}$$

$$V_{inv} = \begin{bmatrix} U_{ai} & U_{bi} & U_{ci} \end{bmatrix}^T \tag{3.8}$$

$$\begin{cases} Uai = (Ta - T_n)Vdc \\ Ubi = (Tb - T_n)Vdc \\ Uci = (Tc - T_n)Vdc \end{cases} \tag{3.9}$$

$$V_{grid} = \begin{bmatrix} V_{ga} & V_{gb} & V_{gc} & \end{bmatrix}^T \tag{3.10}$$

$$I_1 = \begin{bmatrix} I_{1a} & I_{1b} & I_{1c} & \end{bmatrix}^T \tag{3.11}$$

$$I_2 = \begin{bmatrix} I_{2a} & I_{2b} & I_{2c} & \end{bmatrix}^T \tag{3.12}$$

The mathematical model of the entire system is represented in continuous form as:

$$\begin{cases} L_eq.\frac{dI_1}{dt} + R_fC_f.\frac{dV_c}{dt} = V_{inv} - V_{cap} \\ C_f.\frac{dV_c}{dt} = I_1 - I_2 \\ R_fC_f.\frac{dV_c}{dt} - L_2\frac{dI_1}{dt} = V_{grid} - V_{cap} \end{cases} \tag{3.13}$$

These equations can be represented is the general form as

$$\begin{cases} \frac{dx}{dt} = Ax + Bu + Evgrid \\ y = Cx \end{cases} \tag{3.14}$$

where

$$x = \begin{bmatrix} I_d & I_q & V_d & V_q \end{bmatrix}^T \tag{3.15}$$

In particular, digital implementation of model predictive control requires discrete time model. Thus, the discrete form is written as follows:

$$\begin{cases} x_m(k+1) = A_m x_m(k) + B_m u(k) + E_m V_{grid}(K) \\ \hfill y_k = C_m x_m(k) \end{cases} \tag{3.16}$$

where u(k) is the manipulated variable or input variable; y(k) is the process output; and x_m is the state variable vector. It can be assumed that the input u(k) cannot affect the output y(k)at the same time. Besides, $A_m = e^{AT_s}$,$B_m = \int_0 T_s e^{At} B dt$ and $E_m = \int_0 T_s e^{At} E dt$

$$\begin{cases} P_{inv} = \frac{3}{2}(V_d I_d + V_q I_q) \\ Q_{inv} = \frac{3}{2}(-V_d I_q + V_q I_d) \end{cases} \tag{3.17}$$

The derivative of active and reactive powers with respect to time can be written as

$$\begin{bmatrix} \frac{P_{inv}}{dt} \\ \frac{Q_{inv}}{dt} \end{bmatrix} = \frac{2}{3} \begin{bmatrix} \frac{V_d}{dt} & \frac{V_q}{dt} \\ \frac{V_q}{dt} & -\frac{V_d}{dt} \end{bmatrix} \begin{bmatrix} I_d \\ I_q \end{bmatrix} + \frac{2}{3} \begin{bmatrix} V_d & V_q \\ V_q & -V_d \end{bmatrix} \begin{bmatrix} \frac{I_d}{dt} \\ \frac{I_q}{dt} \end{bmatrix} \tag{3.18}$$

The dynamics of diesel driven generator and its associated to its automatic voltage regulator (AVR) is modelled as follows

$$P_{ddg} = V_{dd} I_{dd} \tag{3.19}$$

$$Q_{ddg} = V_{dq} I_{dq} \tag{3.20}$$

The derivative of active and reactive powers of diesel driven generator with respect to time can be written as

$$\begin{bmatrix} \frac{P_{ddg}}{dt} \\ \frac{Q_{ddg}}{dt} \end{bmatrix} = \begin{bmatrix} \frac{V_{dd}}{dt} \\ \frac{V_{dq}}{dt} \end{bmatrix} \begin{bmatrix} I_{dd} \\ I_{dq} \end{bmatrix} + \begin{bmatrix} V_{dd} \\ V_{dq} \end{bmatrix} \begin{bmatrix} \frac{I_{dd}}{dt} \\ \frac{I_{dq}}{dt} \end{bmatrix} \tag{3.21}$$

The cost function subject to minimization can be formed as:

$$minJ(k) = t_s(\omega_{gr} \sum_{k=1}^{N} t(t) P_{grid}(k) + \omega_{ddg} \sum_{k=1}^{N} C_f(aP_{ddg}^2 + bP_{ddg} + c) - \omega_{pv} F_i \sum_{k=1}^{N} P_{pv}(k)) \tag{3.22}$$

where, t(t)is the time of use electricity tariff in [Ksh/Kwh], C_f is the fuel cost of the diesel driven generator [Ksh/litre], F_i is feed-in tariff of renewable energy from solar PV in [Ksh/kWh], t_s is the sampling time(h), N is the sampling interval , and ω_{pv}, ω_{dg} and ω_{gr} are weighting factors.

$$J(k+1) = (P_{ref(k+1)} - P_{out}(k+1)) + (Q_{ref}(k+1) - Q_{out}(k+1)) \tag{3.23}$$

Model predictive algorithm

Model predictive control (MPC) algorithm is based on the discrete system model, which is used to predict system response during the next sampling period. The main objective of the model predictive control (MPC) algorithm is to control the active and reactive power through the control of the d and q grid current components. Controlling the active and reactive power efficiently through this method can improve the performances optimization and operational efficiency in both grid connected and off grid modes. The advantage of this method is that it has faster computational time by enhancing the steady-state and transient state control problems separately. To achieve this, mathematical model of the plant have to be determined. This helps to determine the state-space equations of the system.

Afterwards, optimal control parameters and gains for controllers are determined to optimally balance powers between photovoltaic system side and grid side on one hand, and photovoltaic-diesel generator in case of outage of grid, on the other hand. Besides, Solving this problem consists on determining the objective function in order to figure out the optimal parameters and gains of the controller. In grid-connected photovoltaic system, the optimal control will minimize the grid energy consumption needed and to miximize the power from PV system. Conversely, in autonomous grid mode, the objective function is built to penalize the use of diesel energy generation and the use of PV energy is miximized as well.[8]. As a result, the weighing factors ω_{pv}, ω_{dg} and ω_{gr} are adjusted based on the desired effects. In general the cost function is represented as follows:

$$minJ(k) = min(Y(k) - R(k))^T (Y(k) - R(k)) \tag{3.24}$$

The system is subject to some constraints, some are more important than others. Furthermore, the active power should be balanced and the reactive power should be zero at any time t. The require load demand must be satisfied by the total power sources. This can be expressed as:

$$\begin{cases} P_{load,t} = P_{grid,t} + P_{pv,t} + P_{ddg,t} \\ Q^{*} = 0 \end{cases} \tag{3.25}$$

Photovoltaic power source is to be controlled in the range of zero to their rated power, but grid power that is to be controlled as infinity node to supply load demand whenever renewable energy is insufficient. For the diesel generator, the minimum output power recommended by the manufacturer is 30 % of the rated power. Moreover, if the diesel produces power continuously, the produced power cannot exceed 90% of its rated power.

$$0 \leq P_{pv}(k) \leq P_{pv}^{max}(k) \tag{3.26}$$

$$0 \leq P_{ddg}(k) \leq P_{ddg}^{rated}(k) \tag{3.27}$$

$$-\infty \leq P_{grid} \leq +\infty \tag{3.28}$$

Model predictive control (MPC) technique allows to explicitly handle optimality and system constraints. The constraints of the system are modelled with respect to its practical limitations and are expressed in a compact form as illustrated above.

From the control point of view, three-phase grid-connected photovoltaic -diesel generator system is considered as Multi-Input Multi-Output (MIMO) system whose voltages and currents are the two basic signals. Most specifically, voltages, currents and phase angle must be measured. The significant advantage of the Model predictive control (MPC) over other control strategy is the ability of optimizing multi-variable and deal with constraints.

Moreover, an useful unit in three-phase system to synchronize the photovoltaic to the utility grid is a phase-locked loop (PLL). Phase-locked loop (PLL) is the most suitable method to track the phase angle θ of the input signal quickly and accurately [12]. However, the phase-locked loop (PLL) as a nonlinear part, usually degrades the output performance of system. Consequently, it is good to propose a model predictive control without these modules, which can make the control algorithm simpler and reduce cost of the design. With the linear state-space, constraints and the objective function optimal model predictive control(OMPC) for

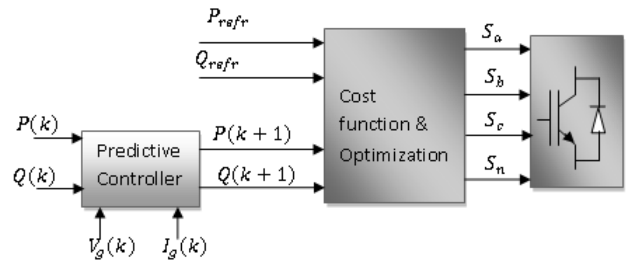

Figure 3.4: structure of model predictive control

MIMO linear system can be applied. Block diagram of model predictive control is displayed figure 3.4. The control structure of model predictive control considers above to the grid connected converter model in the d-q synchronous reference frame to forecast possible future behaviour of the grid.

1. calculate MPC gain

$$Y = Fx(k) + \Phi\Delta U \tag{3.29}$$

where, $F=\begin{bmatrix} CA \\ CA^2 \\ . \\ . \\ . \\ CA^{Np} \end{bmatrix}$; $\Phi = \begin{bmatrix} CB & 0 & . \; . & . & 0 \\ CAB & CB & . \; . & . & 0 \\ . & . & . \; . & . & . \\ . & . & . \; . & . & . \\ . & . & . \; . & . & . \\ CA^{Np-1}B & CA^{Np-2}B & . \; . & CA^{Np-Nc}B & \end{bmatrix}$

$$E = \Phi^T * \Phi \tag{3.30}$$

$$H = (Fx(k) - R(k))^T * \Phi \tag{3.31}$$

2. Express the output vector with respect to input series

$$minJ(k) = min(Y(k) - R(k))^T (Y(k) - R(k)) \tag{3.32}$$

3. Find optimal inputs u(k) such that the objective function is minimize and constraints are satisfied

$$minU^T(k)EU(k) + 2HU(k) \tag{3.33}$$

4. Calculate the receding horizon control although the optimal parameter vector ΔU contains the controls $\Delta u(k)$, $\Delta u(k+1)$, $\Delta u(k+2)$,..., $\Delta u(k+Nc-1)$.
   where Nc is the control horizon dictating the number of parameters used to capture the future control strategy. on one hand, receding horizon control principle, it is intended to implement the first sample of the sequence.on the other hand this implementation of receding horizon control is capable of addressing disturbance.

5. Set k = k + 1, and update system states, inputs and outputs with the control u(k) and state-space equations . Repeat steps (4) until k reaches its desired value [27].

In fact,this appropriate optimal control model will be carried out in MATLAB/SIMULINK. Various operating conditions will be considered to validate the optimal operation and control strategy.

3.0.3 Estimation of payback period

The payback period (PBP) is the time that is expected before an investment will be returned in the form of income [15].

$$PBP = \frac{Cost_{NPC}}{Cost_{NCF}} \tag{3.34}$$

where $Cost_{NPC}$ the total net present cost and $Cost_{NCF}$ represent the total net present cost and total net cash flow respectively.

Chapter 4

Expected Result

Owing to high rate of energy demand and periodic outage, three-phase grid connected photovoltaic with diesel generator backup system in medium scale must play a major role in many developing countries specially in Kenya. Therefore, in this research, it is expected to optimize the design and control strategy of the three-phase grid tied photovoltaic-diesel generator backup system. As consequence, the novel design optimal design and control scheme strategy should improve the performances of the overall system in both grid connected and autonomous grid modes.

In order to obtain successful result, Jomo Kenyatta University of Agriculture and Technology (JKUAT) is chosen as a case study for its real load profiles. As this system is not yet to Jomo Kenyatta University of Agriculture and Technology (JKUAT); First of all, it is intended to optimally designed the system in order to improve the economic benefit, reliability and reduce the greenhouse gas emissions.

Secondly, the newly optimum operation control structure must be capable to guarantee operational efficiency by considering system constraints, unbalanced and no-linear loads, and time of use in Jomo Kenyatta University of Agriculture and Technology (JKUAT). In particular, the cost function is constructed to penalize the use of diesel generator while encourage the use of photovoltaic array system. It is intended that the optimal control strategy can lead to the savings of energy and costs in the microgrid. As a result, a well managed grid connected photovoltaic with diesel driven generator backup can assist the end-user in substantially reducing electricity cost, enhance the security and efficiency performances while increasing the reliability. The simulation is done on one hand by not considering the maximum power point tracking and on the other hand by implementing the most commercial used maximum power point tracking (MPPT) based on perturb-observe (P-O).

Lastly, It is important to assess the feasibility of implementing an optimal design and control of grid connected photovoltaic with generator backup, not only in terms of operation cost, reliability and environmental benefits, but also based on payback period. Payback period must therefore be determined. This is time for the cost effectiveness of implementing this project and harvesting solar energy based on comprehensive consideration of various cost and revenue components at Jomo Kenyatta University of Agriculture and Technology (JKUAT). The expectation is to assess how the optimal design and control strategy scheme for appropriated time of use and feed in tariff in Kenya can significantly reduce the payback period owing to huge revenues from solar energy sales benefits.

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